

LAB 3:

Aim: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

1. Identify all entity types and entity sets.

The University Research Management System consists of six main entity types and their corresponding entity sets: Researcher, Department, Research Project, Student, Publication, and Project Assistant. The Researcher entity set represents all researchers in the university, while the Department entity set represents all departments. The Research Project entity set represents all research projects undertaken by the university, and the Student entity set represents all students working under researchers. The Publication entity set represents all publications produced from research projects. The Project Assistant entity set represents all assistants associated with research projects. Among these, Researcher, Department, Research Project, Student, and Publication are strong entity sets, whereas Project Assistant is a weak entity set because it is identified in the context of its associated Research Project.

2. Identify strong and weak entity sets.

- **Strong entities**
- Researcher
- Department
- Research Project
- Student
- Publication
- **Weak entity**
- Project Assistant

3. Identify the owner entity and partial key of the weak entity.

Weak Entity: Project Assistant

Owner entity: Research Project

Partial key: Assistant_Name

4. Identify simple attributes.

Simple attributes cannot be meaningfully divided into smaller attributes.

- Researcher: Researcher_ID, Name, Date_of_Birth, Email
- Department: Department_ID, Department_Name, Location
- Project: Project_ID, Project_Title, Start_Date, End_Date, Budget
- Student: Student_ID, Name, Email, Date_of_Birth
- Publication: Publication_ID, Title, Publication_Date, DOI, Publisher
- Assistant: Assistant_Name, Role

5. Identify composite attributes.

Researcher Address

Address is composite:

Address

- Street
- City
- State
- PIN_Code

6. Identify single-valued attributes.

Single-Valued Attributes

Examples:

- Researcher_ID
- Name
- Date_of_Birth
- Email
- Department_ID
- Department_Name
- Location
- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget
- Student_ID
- Publication_ID
- DOI
- Publisher
- Assistant_Name

- Role

Each occurrence of the entity has one value for these attributes.

7. Identify multi-valued attributes.

There are two important multi-valued attributes:

Phone_Number and Keywords

8. Identify stored attributes.

Stored Attributes

These are actually maintained/stored in the database.

Examples:

- Date_of_Birth
- Start_Date
- End_Date
- Budget
- Publication_Date
- Name
- Email

9. Identify derived attributes.

Derived Attribute

Age

Age is derived from Date_of_Birth.

10. Identify key attributes.

The key attributes of the entities in the University Research Management System are as follows. Researcher is identified uniquely by Researcher_ID, which is its primary key. Department is identified by Department_ID, which is its primary key. Research Project is identified by Project_ID, which is its primary key. Student is identified by Student_ID, which is its primary key. Publication is identified by Publication_ID, which is its primary key. Project Assistant is a weak entity and is identified using Assistant_Name as its partial key. Since the same assistant name can occur in different projects, the complete key for Project Assistant is the composite key (Project_ID, Assistant_Name), where Project_ID is obtained from its owner entity, Research Project.

11. Identify candidate keys and select appropriate primary keys.

Entity	Candidate Key	Selected Primary Key
Researcher	Researcher_ID	Researcher_ID
Department	Department_ID	Department_ID
Research Project	Project_ID	Project_ID
Student	Student_ID	Student_ID
Publication	Publication_ID	Publication_ID
Project Assistant	(Project_ID, Assistant_Name)	(Project_ID, Assistant_Name)

12. Identify at least one composite key.

Assitant_Name

13. Identify appropriate foreign keys after converting the ER model into relations.

DEPARTMENT

DEPARTMENT(Department_ID, Department_Name, Location)

- PK: Department_ID

RESEARCHER

RESEARCHER(Researcher_ID, Name, Date_of_Birth, Email, Department_ID)

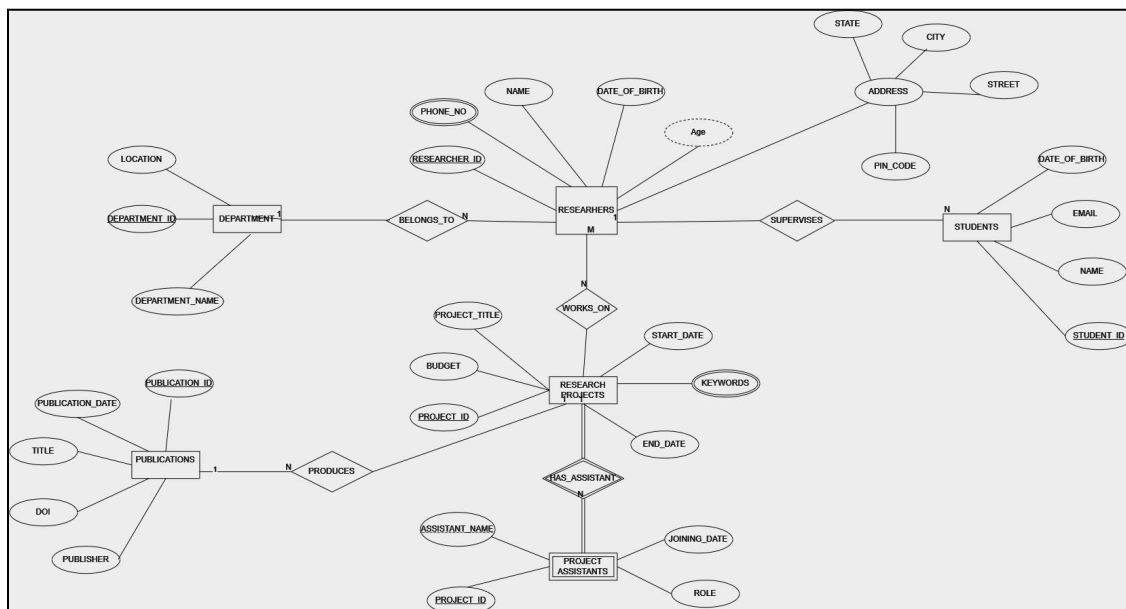
- PK: Researcher_ID
- FK: Department_ID → DEPARTMENT

14. Identify all major relationships between the entities.

	Relationship	Entities Involved	Cardinality	Meaning
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	BELONGS_TO	Department ↔ Researcher	1 : N	One department has many researchers; each researcher belongs to one department.
	WORKS_ON	Researcher ↔ Research Project	M : N	A researcher can work on many projects, and a project can involve many researchers.
	SUPERVISES	Researcher ↔ Student	1 : N	One researcher can supervise many students; each student works under one researcher.
	PRODUCES	Research Project ↔ Publication	1 : N	One project can produce many publications; each publication belongs to one project.
	HAS_ASSISTANT	Research Project ↔ Project Assistant	1 : N	One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity.

15. Draw the complete ER diagram using standard ER notation.



16. Prepare a table showing each attribute and its attribute type and key status.

Entity	Attribute	Attribute Type	Value	Key Status
Researcher	Researcher_ID	Simple	Single	Primary Key
Researcher	Name	Simple	Single	Non-key
Researcher	Date_of_Birth	Simple	Single	Non-key, Stored
Researcher	Age	Simple/Derived	Single	Derived
Researcher	Email	Simple	Single	Non-key
Researcher	Phone_Number	Simple	Multi-valued	Non-key
Researcher	Address	Composite	Single	Non-key
Researcher	Street	Simple component	Single	Non-key
Researcher	City	Simple component	Single	Non-key
Researcher	State	Simple component	Single	Non-key
Researcher	PIN_Code	Simple component	Single	Non-key
Department	Department_ID	Simple	Single	PrimaryKey
Department	Department_Name	Simple	Single	Non-key
Department	Location	Simple	Single	Non-key
Research Project	Project_ID	Simple	Single	Primary Key
Research Project	Project_Title	Simple	Single	Non-key
Research Project	Start_Date	Simple	Single	Stored
Research Project	End_Date	Simple	Single	Stored
Research Project	Budget	Simple	Single	Stored
Research Project	Keyword	Simple	Multi-valued	Non-key
Student	Student_ID	Simple	Single	Primary Key
Student	Name	Simple	Single	Non-key

Student	Email	Simple	Single	Non-key
Student	Date_of_Birth	Simple	Single	Stored
Publication	Publication_ID	Simple	Single	PrimaryKey
Publication	Title	Simple	Single	Non-key
Publication	Publication_Date	Simple	Single	Stored
Publication	DOI	Simple	Single	Non-key
Publication	Publisher	Simple	Single	Non-key
Project Assistant	AssistantName	Simple	Single	Partial Key
Project Assistant	Role	Simple	Single	Non-key
